REFINING AN FPQC COVERING OF AN AFFINE SCHEME BY A STANDARD FPQC COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [3367, 3376] and proof(s) [[3378, 3380]]; this spec covers the complete environment instead (statement 3367-3376, proof span(s) [(3378, 3380)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Lemma 0.1. *Let T be an affine scheme. Let $\{T_i \to T\}_{i \in I}$ be an fpqc covering of T. Then there exists an fpqc covering $\{U_j \to T\}_{j=1,\ldots,n}$ which is a refinement of $\{T_i \to T\}_{i \in I}$ such that each U_j is an affine scheme. Moreover, we may choose each U_j to be open affine in one of the T_i.*

Proof. This follows directly from the definition. □

Date: stacks-project:topologies.tex:local-snapshot-20261012; GNU FDL 1.2 or later.